

Thermally Stable Crystalline Coatings for Corrosive Media

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Abstract

Work in materials research has long relied on assumptions we test directly here. Neither of the alternative formulations improved on the simple additive model in our setting. Taken together, these findings argue for treating the two regimes separately in future analyses. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

1. Introduction

The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Neither of the alternative formulations improved on the simple additive model in our setting. The measurements reported here were collected over twelve independent trials under controlled conditions.

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We release the processing scripts and intermediate artefacts to support independent verification. Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Longer observation windows narrow the confidence interval but do not shift the central estimate. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The reduced model retains most of the explanatory power at a fraction of the computational cost. Sensitivity to the smoothing window proved modest across the range we examined.

2. Related Work

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3. Materials and Methods

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4. Experimental Setup

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5. Results

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6. Discussion

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